

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Logarithmic Secant Integral

Evaluate the definite integral:

$$\int_0^{\frac{\pi}{2}} \frac{\ln(\sec(x))}{\sin(x)} dx$$

Solution: First, rewrite the integrand by noting that $\ln(\sec(x)) = -\ln(\cos(x))$:

$$I = \int_0^{\frac{\pi}{2}} \frac{-\ln(\cos(x))}{\sin(x)} dx$$

Apply the substitution $u = \cos(x)$. The differential is $du = -\sin(x) dx$, which implies $dx = -\frac{du}{\sin(x)}$. The limits of integration change from $x = 0 \Rightarrow u = 1$ to $x = \frac{\pi}{2} \Rightarrow u = 0$. Substituting these into the integral gives:

$$I = \int_1^0 \frac{-\ln(u)}{\sin(x)} \left(-\frac{du}{\sin(x)}\right) = \int_0^1 \frac{-\ln(u)}{1-u^2} du$$

This is a well-known standard integral. We can evaluate it by expanding $\frac{1}{1-u^2}$ into a geometric series $\sum_{k=0}^{\infty} u^{2k}$:

$$I = -\sum_{k=0}^{\infty} \int_0^1 u^{2k} \ln(u) du$$

Using integration by parts, it is easily shown that $\int_0^1 u^n \ln(u) du = -\frac{1}{(n+1)^2}$. Thus:

$$I = \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2}$$

This sum is formed by the odd terms of the Basel problem series. Knowing that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, we subtract the even terms $\frac{1}{4} \frac{\pi^2}{6}$ to find the sum of the odd terms:

$$I = \frac{\pi^2}{6} - \frac{\pi^2}{24} = \frac{\pi^2}{8}$$

1.2 Hard Integral

Logarithmic Rational Integral

Evaluate the definite integral:

$$\int_0^1 \frac{\ln(x) \ln(1-x^2)}{x^2} dx$$

Solution: We start by expanding the logarithmic term $\ln(1 - x^2)$ into its Taylor series around $x = 0$:

$$\ln(1 - x^2) = - \sum_{n=1}^{\infty} \frac{x^{2n}}{n}$$

Substitute this series back into the integral. Since the convergence is uniform on the interval $(0, 1)$, we can interchange the sum and the integral:

$$I = \int_0^1 \frac{\ln(x)}{x^2} \left(- \sum_{n=1}^{\infty} \frac{x^{2n}}{n} \right) dx = - \sum_{n=1}^{\infty} \frac{1}{n} \int_0^1 x^{2n-2} \ln(x) dx$$

Evaluate the integral $\int_0^1 x^{2n-2} \ln(x) dx$ using integration by parts, which gives $-\frac{1}{(2n-1)^2}$.

$$I = \sum_{n=1}^{\infty} \frac{1}{n(2n-1)^2}$$

To evaluate this sum, use partial fraction decomposition on the general term:

$$\frac{1}{n(2n-1)^2} = \frac{1}{n} - \frac{2}{2n-1} + \frac{2}{(2n-1)^2}$$

Split the sum into two parts:

$$I = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{2}{2n-1} \right) + 2 \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$$

The second sum is the sum of the reciprocals of odd squares, which we evaluated in the previous problem as $\frac{\pi^2}{8}$. Thus, $2 \left(\frac{\pi^2}{8} \right) = \frac{\pi^2}{4}$. The first sum can be written as a limit of partial sums:

$$\sum_{n=1}^N \left(\frac{1}{n} - \frac{2}{2n-1} \right) = \sum_{n=1}^N \frac{1}{n} - 2 \sum_{k=1}^N \frac{1}{2k-1} = -2 \ln(2)$$

Putting it all together, the exact value of the integral is:

$$I = \frac{\pi^2}{4} - 2 \ln(2)$$

Differentials

2.1 Medium Differential

Linear Second-Order Non-Homogeneous ODE

Solve the second-order linear non-homogeneous differential equation:

$$y'' - 5y' + 6y = 12x$$

Given the initial conditions $y(0) = 2$ and $y'(0) = 5$, find the exact value of $y(1)$.

Solution: First, find the complementary solution $y_c(x)$ by solving the homogeneous equation $y'' - 5y' + 6y = 0$. The characteristic equation is:

$$r^2 - 5r + 6 = 0 \implies (r - 2)(r - 3) = 0$$

Thus, $r = 2$ or $r = 3$. The complementary solution is:

$$y_c(x) = C_1 e^{2x} + C_2 e^{3x}$$

Next, find a particular solution $y_p(x)$ using the method of undetermined coefficients. Since the right side is a linear polynomial, guess $y_p(x) = Ax + B$. The derivatives are $y'_p = A$ and $y''_p = 0$. Substitute these into the ODE:

$$0 - 5(A) + 6(Ax + B) = 12x \implies 6Ax + (6B - 5A) = 12x$$

Equating coefficients yields:

$$6A = 12 \implies A = 2$$

$$6B - 5A = 0 \implies 6B - 10 = 0 \implies B = \frac{5}{3}$$

The general solution is $y(x) = C_1 e^{2x} + C_2 e^{3x} + 2x + \frac{5}{3}$. Now, apply the initial conditions to find C_1 and C_2 .

$$y(0) = C_1 + C_2 + \frac{5}{3} = 2 \implies C_1 + C_2 = \frac{1}{3}$$

The derivative is $y'(x) = 2C_1 e^{2x} + 3C_2 e^{3x} + 2$.

$$y'(0) = 2C_1 + 3C_2 + 2 = 5 \implies 2C_1 + 3C_2 = 3$$

Solving this system, multiply the first equation by 2 to get $2C_1 + 2C_2 = \frac{2}{3}$. Subtracting this from the second equation gives $C_2 = \frac{7}{3}$. Substituting C_2 back gives $C_1 = \frac{1}{3} - \frac{7}{3} = -2$. The specific solution is:

$$y(x) = -2e^{2x} + \frac{7}{3}e^{3x} + 2x + \frac{5}{3}$$

Finally, evaluate at $x = 1$:

$$y(1) = -2e^2 + \frac{7}{3}e^3 + \frac{11}{3}$$

2.2 Hard Differential

Euler-Lagrange Equation

Using Euler-Lagrange equations, find a function $y(x)$ such that the functional

$$S = \int_0^2 \left(x^2 + y(x) \left(\frac{dy}{dx} \right)^2 \right) dx$$

is at its stationary point. Knowing that $y(0) = 0$ and $y(2) = 2$, compute the value of the functional at its stationary point.

Solution: Let the Lagrangian be $L(x, y, y') = x^2 + y(y')^2$. The Euler-Lagrange equation is:

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \left(\frac{\partial L}{\partial y'} \right) = 0$$

Compute the partial derivatives:

$$\frac{\partial L}{\partial y} = (y')^2, \quad \frac{\partial L}{\partial y'} = 2yy'$$

Substitute these into the equation:

$$\begin{aligned} (y')^2 - \frac{d}{dx}(2yy') &= 0 \implies (y')^2 - (2(y')^2 + 2yy'') = 0 \\ -(y')^2 - 2yy'' &= 0 \implies 2yy'' + (y')^2 = 0 \end{aligned}$$

To solve this nonlinear ODE, multiply both sides by y' :

$$2yy'y'' + (y')^3 = 0 \implies \frac{d}{dx} (y(y')^2) = 0$$

Integrating gives:

$$y(y')^2 = C_1 \implies y' = \sqrt{\frac{C_1}{y}}$$

Separate the variables and integrate:

$$\sqrt{y} dy = \sqrt{C_1} dx \implies \frac{2}{3} y^{3/2} = \sqrt{C_1} x + C_2$$

Using $y(0) = 0$, we find $C_2 = 0$. Thus, $y(x)^{3/2} = \frac{3}{2} \sqrt{C_1} x$. Let $K = \frac{3}{2} \sqrt{C_1}$, so $y(x) = (Kx)^{2/3}$. Use the boundary condition $y(2) = 2$ to find K :

$$(2K)^{2/3} = 2 \implies 2K = 2^{3/2} \implies K = \sqrt{2}$$

The function is $y(x) = (\sqrt{2}x)^{2/3} = 2^{1/3}x^{2/3}$. Next, compute the derivative y' :

$$y' = \frac{2}{3} \cdot 2^{1/3} x^{-1/3}$$

Calculate the term $y(y')^2$:

$$y(y')^2 = \left(2^{1/3}x^{2/3}\right) \left(\frac{4}{9} \cdot 2^{2/3}x^{-2/3}\right) = \frac{8}{9}$$

Now evaluate the functional S :

$$S = \int_0^2 \left(x^2 + \frac{8}{9}\right) dx = \left[\frac{x^3}{3} + \frac{8}{9}x\right]_0^2 = \frac{8}{3} + \frac{16}{9} = \frac{24+16}{9} = \frac{40}{9}$$

Derivatives

3.1 Medium Derivative

Finite Product Derivative

Consider the product $f(x) = \prod_{n=1}^{50} (1 + nx)$. Find $f'(0)$.

Solution: To find the derivative of a large product, apply logarithmic differentiation. First, take the natural logarithm of both sides:

$$\ln(f(x)) = \sum_{n=1}^{50} \ln(1 + nx)$$

Differentiate both sides with respect to x :

$$\frac{f'(x)}{f(x)} = \sum_{n=1}^{50} \frac{n}{1 + nx}$$

To find $f'(0)$, we must evaluate both $f(0)$ and the sum at $x = 0$. For $f(0)$, substitute $x = 0$ into the original product:

$$f(0) = \prod_{n=1}^{50} (1 + 0) = 1$$

Now substitute $x = 0$ into the differentiated equation:

$$\frac{f'(0)}{1} = \sum_{n=1}^{50} \frac{n}{1 + 0} = \sum_{n=1}^{50} n$$

This is an arithmetic progression of integers from 1 to 50. Using the sum formula for the first N positive integers, $S = \frac{N(N+1)}{2}$:

$$f'(0) = \frac{50 \times 51}{2} = 25 \times 51 = 1275$$

3.2 Hard Derivative

Mixed Partial Derivative of Special Functions

Find $\frac{\partial^2}{\partial a \partial b} \left(\frac{B(a,b)}{\zeta(a)\zeta(b)} \right)$ at $a = 2, b = 2$.

Solution: Let $F(a, b) = \frac{B(a,b)}{\zeta(a)\zeta(b)} = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)\zeta(a)\zeta(b)}$. We are looking for $F_{ab}(2, 2)$. Take the natural logarithm to simplify differentiation:

$$\ln(F) = \ln \Gamma(a) + \ln \Gamma(b) - \ln \Gamma(a+b) - \ln \zeta(a) - \ln \zeta(b)$$

Differentiate with respect to b :

$$\frac{\partial \ln F}{\partial b} = \psi(b) - \psi(a+b) - \frac{\zeta'(b)}{\zeta(b)}$$

where $\psi(z)$ is the digamma function. Now differentiate with respect to a :

$$\frac{\partial^2 \ln F}{\partial a \partial b} = -\psi'(a+b)$$

Using the identity $\frac{\partial^2 \ln F}{\partial a \partial b} = \frac{F_{ab}F - F_a F_b}{F^2}$, we solve for F_{ab} :

$$F_{ab} = F \left(\frac{\partial^2 \ln F}{\partial a \partial b} + \frac{\partial \ln F}{\partial a} \frac{\partial \ln F}{\partial b} \right)$$

Evaluate the terms at $a = 2, b = 2$:

$$F(2,2) = \frac{\Gamma(2)\Gamma(2)}{\Gamma(4)\zeta(2)^2} = \frac{1 \cdot 1}{6(\pi^2/6)^2} = \frac{6}{\pi^4}$$

The first derivatives at $(2,2)$ are symmetric:

$$\frac{\partial \ln F}{\partial a} = \frac{\partial \ln F}{\partial b} = \psi(2) - \psi(4) - \frac{\zeta'(2)}{\zeta(2)}$$

Using the property $\psi(n+1) = -\gamma + H_n$, we find $\psi(4) - \psi(2) = H_3 - H_1 = \left(1 + \frac{1}{2} + \frac{1}{3}\right) - 1 = \frac{5}{6}$. Thus, the first derivative simplifies to $-\frac{5}{6} - \frac{6\zeta'(2)}{\pi^2}$. The second mixed logarithmic derivative requires $\psi'(4)$:

$$\psi'(4) = \sum_{k=0}^{\infty} \frac{1}{(4+k)^2} = \frac{\pi^2}{6} - 1 - \frac{1}{4} - \frac{1}{9} = \frac{\pi^2}{6} - \frac{49}{36}$$

So $\frac{\partial^2 \ln F}{\partial a \partial b}(2,2) = \frac{49}{36} - \frac{\pi^2}{6}$. Substitute everything back into F_{ab} :

$$F_{ab}(2,2) = \frac{6}{\pi^4} \left(\frac{49}{36} - \frac{\pi^2}{6} + \left(\frac{5}{6} + \frac{6\zeta'(2)}{\pi^2} \right)^2 \right)$$

Physics & Engineering

4.1 Electromagnetism

Magnetic Force on a Wire

A rigid wire bent into the shape of a semi-circle of radius $R = 0.5$ m carries a steady current of $I = 10$ A. It is placed in a uniform magnetic field of magnitude $B = 0.2$ T directed perpendicular to the plane of the semi-circle. Find the magnitude of the total magnetic force acting on the wire in newtons (N).

Solution: The magnetic force $d\vec{F}$ acting on an infinitesimal element $d\vec{l}$ of a wire carrying current I in a magnetic field $\vec{B}$ is given by the Biot-Savart Law force equivalent:

$$d\vec{F} = I(d\vec{l} \times \vec{B})$$

When the external magnetic field $\vec{B}$ is uniform, the total force $\vec{F}$ on any shape of wire can be found by integrating over its entire length:

$$\vec{F} = \int I(d\vec{l} \times \vec{B}) = I \left(\int d\vec{l} \right) \times \vec{B}$$

The integral $\int d\vec{l}$ is simply the net displacement vector $\vec{L}$ connecting the start point to the end point of the wire. For a semi-circle of radius R , the displacement vector connecting the two ends is a straight line across the diameter. Thus, its length is:

$$L = 2R = 2(0.5 \text{ m}) = 1.0 \text{ m}$$

Since the magnetic field $\vec{B}$ is perpendicular to the plane of the semi-circle, it is also perfectly perpendicular to the displacement vector $\vec{L}$. The magnitude of the force is then simply:

$$F = ILB = (10 \text{ A})(1.0 \text{ m})(0.2 \text{ T}) = 2.0 \text{ N}$$

4.2 Mechanics

Block on a Rough Circular Track

A block of mass m is released from rest at the top of a quarter-circular track of unit radius $R = 1$ m. The track is rough with a coefficient of kinetic friction $\mu = 0.5$. Find the speed of the block when it reaches the bottom of the track. Take $g = 10 \text{ m s}^{-2}$.

Solution: Let θ be the angle of the block measured from the top vertical position. The forces acting on the block radially are the normal force N and the radial component of gravity $mg \sin \theta$. Using Newton's second law for uniform circular motion:

$$N - mg \sin \theta = \frac{mv^2}{R} \implies N = mg \sin \theta + \frac{mv^2}{R}$$

The frictional force is $f_k = \mu N$. The tangential equation of motion is:

$$mg \cos \theta - f_k = m \frac{dv}{dt} = \frac{mv}{R} \frac{dv}{d\theta}$$

Substitute the friction force into the tangential equation:

$$mg \cos \theta - \mu \left(mg \sin \theta + \frac{mv^2}{R} \right) = \frac{m}{2R} \frac{d(v^2)}{d\theta}$$

Let $u = v^2$. The equation simplifies to a first-order linear ODE:

$$\frac{du}{d\theta} + 2\mu u = 2gR(\cos \theta - \mu \sin \theta)$$

Multiply by the integrating factor $e^{2\mu\theta}$:

$$\frac{d}{d\theta} (ue^{2\mu\theta}) = 2gRe^{2\mu\theta}(\cos \theta - \mu \sin \theta)$$

Integrate from $\theta = 0$ to $\theta = \frac{\pi}{2}$:

$$u \left(\frac{\pi}{2} \right) e^{\mu\pi} - 0 = \frac{2gR}{4\mu^2 + 1} \left[e^{2\mu\theta} (3\mu \cos \theta + (1 - 2\mu^2) \sin \theta) \right]_0^{\pi/2}$$

$$u \left(\frac{\pi}{2} \right) e^{\mu\pi} = \frac{2gR}{4\mu^2 + 1} (e^{\mu\pi}(1 - 2\mu^2) - 3\mu)$$

Divide by $e^{\mu\pi}$ and substitute $R = 1, \mu = 0.5, g = 10$:

$$v^2 = \frac{20}{2} (1 - 2(0.25) - 1.5e^{-\pi/2}) = 10(0.5 - 1.5e^{-\pi/2}) = 5 - 15e^{-\pi/2}$$

The speed v is the square root:

$$v = \sqrt{5 - 15e^{-\pi/2}} \text{ m s}^{-1}$$

4.3 Quantum Mechanics

Bell's Inequality Logical Implication

A source produces pairs of spin-1/2 particles in a maximally entangled singlet state. One particle is sent to Alice and the other to Bob. They independently choose measurement directions and find correlations that violate a Bell inequality. Which ONE of the following conclusions must be true?

Solution: Bell's inequalities are mathematically rigorous bounds derived by assuming a local hidden variable theory. The derivation relies upon two primary physical assumptions: 1. **Locality:** The choice of measurement by one observer cannot instantaneously influence the outcome of the other observer's measurement. 2. **Realism:** Physical properties (like spin along a particular axis) have definite, pre-existing values prior to measurement, independent of observation.

When an experiment (like the one performed by Alice and Bob on the singlet state) yields a statistical correlation that exceeds the bounds set by the inequality, it rigorously proves that the framework of Local Hidden Variable theories cannot correctly describe our reality. Logically, if a derived inequality fails in experiment, the only absolute conclusion that can be drawn is that at least one of the foundational premises required to derive the inequality is incorrect. While this implies nature must be non-local or non-real (or both, or lacks freedom of choice), it does not strictly prove that FTL communication occurred, nor does it definitively prove quantum mechanics is complete. It firmly establishes that:

5. At least one assumption in the derivation is false.